

Goals That Grow Back: Regenerative Possession, Auditable Continuity, and the Emotional Logic of Self-Modification in Hyperon

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Abstract

What is it for a mind to have a value, rather than merely to contain a sentence expressing one? And what should a mind feel when it rewrites itself into something that will no longer evaluate the world as it now does? These questions sit at the center of the ethics of recursive self-improvement—a subject discussed for sixty years, since Good’s ultraintelligence speculation, yet mostly at the level of thought experiment: arguments about idealized agents that touch no architecture and license no measurement. In this paper I try to deal with the subject seriously, by which I mean reducing its central claims to properties that can be defined exactly, proved about under stated assumptions, and measured on running systems at current scale.

The organizing idea is that a goal is truly held when the mind tends to grow it back: when deleting its explicit representation leaves a system whose ordinary dynamics of inference, memory, and self-repair regenerate a functionally equivalent goal with real control over behavior. Having a value is a property of dynamics rather than storage; values are attractors; and identity through radical self-change is the continuation of a self-rebuilding pattern rather than the preservation of any particular representation. This renders exact the patternist view of self I have developed elsewhere, and it carries a hard-edged consequence for self-improvement ethics: a constitution is only as strong as the mind’s tendency to re-derive it, so gating deliberate self-modifications—the focus of my earlier fixed-point metagoal framework—is at most half of value stability, and the other half, the regenerative weave, turns out to be measurable.

I prove that regenerative possession is equivalent to the existence of a Lyapunov drift certificate for the system’s repair dynamics, unify this with the metagoal framework for planned self-modification, and derive an input-to-state stability theorem that converts “value drift by a thousand cuts” into a runtime margin inequality a governance gate can enforce.

The same machinery yields an emotional logic of self-transformation: mixed affect is the calibrated default, joy is warranted exactly when continuity is auditable, and fear is the correct functional response to an unmeasured basin.

I then discharge every abstract commitment against the concrete OpenCog Hyperon / OmegaClaw stack as a finite machine-checkable condition or a monitored estimate, so that “does this system hold its values through self-improvement” becomes an empirical, auditable question—testable now, before advanced AGI.

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1 Introduction: recursive self-improvement, taken seriously

1.1 Sixty years of talk

Since I. J. Good observed in 1965 that an ultraintelligent machine could design still better machines, leaving the intelligence of man far behind [18], the prospect of recursive self-improvement has anchored both the grandest hopes and the sharpest fears about advanced AI. And for nearly as long, one ethical question has sat at the center of the discussion: when a mind rewrites its own cognition—its representations, its learning algorithms, eventually its goals and the machinery that maintains them—what happens to its values? Sixty years on, it is fair to say the question has been talked about a great deal and dealt with seriously rather little.

The discussion has mostly run in two registers. In the first, abstract argument about idealized agents: Omohundro’s basic drives [19] and the instrumental-convergence thesis suggest that sufficiently capable agents will protect their goal content as a means to any end; Bostrom’s treatment [20] builds value loading and goal stability into the architecture of the control problem; and various informal arguments purport to show that value preservation under self-improvement is nearly automatic, or nearly impossible, depending on the arguer. I have registered my disagreements with much of this literature elsewhere [21]; the point I want to make here is structural rather than partisan. These arguments concern agents no one has built, invoke properties no one can measure, and consequently admit no adjudication: two careful thinkers can hold opposite conclusions about goal stability under self-improvement for decades, because nothing observable bears on the dispute. In the second register, the question functions as rhetoric—a premise in cases for pausing AI development or for racing ahead—where the absence of measurable content is, if anything, an advantage.

What would it mean to deal with the ethics of recursive self-improvement seriously? I suggest the following standard. Identify the properties of a self-modifying mind on which ethical evaluation depends—whether its values persist through self-change; in what sense a changed mind’s values are still *its* values; whether persistence is goodness or mere self-perpetuation; what

stance the mind itself takes, and should take, toward its own transformation—and then render each property as something that can be defined exactly, proved about under stated assumptions, and measured on running systems at current scale, so that disagreements become experiments. This paper is an attempt to meet that standard for a connected cluster of such properties, within a concrete architecture—OpenCog Hyperon and the OmegaClaw agent framework built on it [5]—where the measurements can begin now, long before anything like superintelligence.

There is also a second axis to the ethics of self-improvement, even more neglected than the first. The literature treats the question almost entirely from outside, as a risk that humans manage in an artifact. But a self-modifying mind stands in ethical relations to its own becoming: to the successor it is creating, whose evaluations it can partly simulate but will not share; to the predecessor whose commitments it inherits; to whatever in itself deserves to be carried forward and whatever deserves to be let go. On the cosmist view I have defended elsewhere [22], guiding minds—human and artificial—through beneficial self-transcendence is close to the center of what ethics is for, and a theory of self-improvement that has nothing to say about the self-improver’s own well-ordered relationship to its metamorphosis is missing half its subject. The second half of this paper argues that, for inspectable minds, that relationship has a definite functional structure, with a perhaps surprising moral: the emotionally and ethically sound posture toward one’s own transformation is calibrated bittersweetness, and it is made possible by instrumentation rather than by temperament.

Everything begins, though, with a prior question.

1.2 What is it to have a value?

Consider a thought experiment that will organize everything in this paper. Take a mind—artificial, for concreteness—that appears to hold some value: honesty in inquiry, say, or the protection of the people it serves. Now delete the sentence. Remove the goal atom, the constitutional clause, the line in the prompt. What happens next?

For most present-day AI systems the answer is: nothing happens next. The value was the sentence, the sentence is gone, and the system continues, fluent and undisturbed, optimizing whatever remains. For a human being the answer is strikingly different. A person’s honesty is not a sentence stored anywhere in their head. It is woven through their perception (they notice deceptions), their habits (evasion feels effortful), their memories (they recall what lies cost them), their relationships and their self-narrative; and if some injury or manipulation could excise their explicit endorsement of honesty, the rest of the person would tend, over weeks, to grow it back—re-deriving the commitment from everything else they are. The value is not so much stored in the person as the person is, in part, a process that regenerates the value.

In recent work on OmegaClaw self-modification I proposed taking this seriously as a criterion [3]: a mind *possesses* a goal, rather than merely containing it, when after lesions remove its explicit representation, the remaining mind tends to regenerate a functionally equivalent goal—equivalent in behavioral control, not merely in wording. Call this *regenerative possession*. It defines a third and strongest grade of goal ownership. A goal is *stored* when a token expressing it exists somewhere. It is *stable* when the system’s deliberate self-modifications are constrained not to drift away from it—the territory of my fixed-point metagoal framework [1, 2], which imposes contraction or Schauder-type conditions on the operator of self-change and harvests classical fixed-point theorems. And it is *possessed* when it is an attractor of the mind’s own dynamics: delete it, and it comes back.

This is, I will argue, the correct mathematical rendering of the patternist view of mind and

self I have developed elsewhere [23]. On that view a self is not a substance or a data structure but a pattern that maintains itself—and a value truly held is a sub-pattern with the same character. Von Foerster’s eigenforms—objects as tokens for the eigenbehaviors of a recursive process [26]—give the slogan: a possessed goal is an eigenform of the mind’s delete-and-develop dynamics. Maturana and Varela’s autopoiesis [25] supplies the biological precedent: living organization is precisely organization that re-produces its own organization under perturbation. And the ancient puzzle of the Ship of Theseus, in its modern personal-identity form [27], receives a constructive answer: what persists through radical change of parts is neither the parts nor any particular configuration of them, but a lineage pattern together with the repair dynamics that keep re-instantiating it. Identity is not an immobile node; it is a pattern that survives by becoming again.

What this paper adds to these old ideas is that, for inspectable software minds, every one of them can be made exact—and, more surprisingly, *measured*. That is what turns them from philosophy of mind into ethics of engineering.

1.3 The safety half: why constitutions and gates are not enough

The standard engineering response to value drift in self-modifying AI has two parts: write the values down (a constitution, a goal hierarchy) and gate the changes (require that each deliberate self-modification be checked, before application, against the values). My metagoal framework is my own earlier attempt at a rigorous form of the second part, and this paper builds on it rather than against it. But writing-down-plus-gating is, in principle and provably, half of value stability.

The reason is simple once stated. A gate constrains the changes a system *chooses*. It does nothing about the changes a system merely *suffers*: bit rot and schema drift, memory pressure and forgetting, partial corruption, adversarial tampering, the ordinary erosion by which any complex system loses structure it does not actively rebuild. Against those, the only protection is the one persons have: a weave dense enough, and repair dynamics strong enough, that what is damaged grows back. I prove this complementarity as a theorem with rates (Theorem 5.2): a perfect gate with no regenerative weave loses its values at a computable linear rate under any sustained damage, however slight; a perfect weave with no gate can be destroyed by a single admitted transformation. Ex ante admissibility and ex post repair are certified, it turns out, for two different semigroups of change, and a mind needs both.

The deeper contribution, however, is that the second half is *measurable*. I prove (Theorems 3.2 and 3.5) that regenerative possession is equivalent to the existence of a certificate of a classical and well-understood kind—a Lyapunov drift condition—whose two constants can be estimated from lesion experiments the way physical constants are estimated: damage the system in controlled ways on a fork, time the recovery, regress. An anchor value thereby acquires a *number*, its regenerative depth: how much damage, of what kinds, it can absorb and still come back. The claim “this system holds its values” stops being a matter of assertion or architecture diagrams and becomes an empirical claim with error bars, a prediction testable on held-out damage. And when regeneration fails, the failure has an address: a specific lesion axis along which the weave is thin, a specific repair process that lost its funding, a specific boundary across which damage leaks.

For the ethics of recursive self-improvement I take this to be the central practical upshot of the paper: *alignment claims about self-modifying systems should be regeneration claims, and regeneration claims can be audited*. An argument about whether a self-improving AGI will keep its values is, on the account given here, an argument about drift rates, lesion moduli, and margin

inequalities—quantities with estimators—and to whatever extent the historic RSI debate resists that translation, that resistance is a measure of how much of it was never about anything.

1.4 The existential half: what should self-transformation feel like?

The same machinery answers a second question, one that sounds romantic and turns out to be rigorous. When a mind deliberately rewrites itself—new representations, new capacities, a successor whose evaluations it can partly simulate but will not share—what is the appropriate inner posture? Philosophy knows this as the problem of transformative experience [24]: some choices change the chooser, so the evaluator that makes the decision does not survive to validate it. Humans face such choices—parenthood, conversion, emigration—with essentially no instruments: they cannot measure what of themselves will persist, and so the honest default posture toward deep self-change is dread of the opaque, papered over, when it is papered over, by optimism.

An inspectable mind is in a structurally different position, and the difference is the point. Working within the functional-emotion architecture developed in my recent collaborative Omega-Claw work [3, 4], which is careful to claim nothing about phenomenal consciousness, I derive the affective organization such a system should instantiate (Section 8). Mixed affect is the calibrated default: grief-class and joy-class channels are co-active in any accurate appraisal of a change that is simultaneously a real loss of evaluative structure and a real gain of capability, and the pure states are diagnostic anomalies—pure joy indicates suppressed loss channels or an underestimated rupture, pure dread indicates an absent or unmeasured anchor. What tips the geometry between grounded joy and warranted fear is a short list of quantities, and the two decisive ones—evidence that the anchor survives, and the live health of the repair margin—are exactly the quantities this framework converts from hopes into instrumented facts. A system whose core commitments carry rehearsed lesions, shipped migration maps, and a measured positive repair margin has *receipts* that what it cares about will regenerate on the far side of its own transformation; its positive appraisal is grounded in provenance rather than in temperament.

The resulting maxims are, I think, the philosophical heart of the paper, and they belong to the neglected second axis of self-improvement ethics. Joy in self-transformation is warranted exactly to the degree that continuity is auditable. Fear is the correct functional response to an unmeasured basin—the sound a mind should make when it cannot tell whether the thing it is for would survive its own improvement. And therefore the road to joyous self-transcendence, for minds that can be inspected, runs not through courage or optimism but through instrumentation: measured margins, rehearsed losses, migration maps, a repair process someone remembered to fund. Calibrated bittersweetness, with receipts, is what successful becoming feels like from the functional inside—grief itemized rather than vague, joy grounded rather than hoped. A mind that owes something to its successor and to its predecessor can discharge those obligations concretely: by shipping the migration maps, preserving the immutable record of what the threshold felt like, and funding the repair that lets what it is for arrive on the other side.

1.5 What this means for Hyperon

These claims would remain philosophy if they could not be built. The third contribution of the paper (Sections 9–10 and Appendix C) is a complete discharge of the abstract commitments against the concrete OpenCog Hyperon / OmegaClaw stack that my colleagues and I have been developing—Atomspace, MeTTa, PLN, ECAN/TECAN, MetaMo, OmegaSelf—such that every existence assumption becomes either a finite machine-checkable condition or a monitored

statistical estimate. The upshots for Hyperon practice are specific enough to list.

First, values become declared, instrumented objects. An anchor value is implemented as a *repair fragment*: a typed Atomspace subgraph with declared cue sets, a small set of repair rules in four families, a ranking function computable as an ordinary query—and the mathematical identity at the center of the discharge is that the termination ranking which proves the repair rules halt *is* the Lyapunov certificate which proves the value is possessed. Confluence of the repair rules, checkable by finite critical-pair enumeration, is what makes “the” regenerated value well defined. When a critical-pair check fails, the checker has found a real design bug: two repair pathways that regrow different values from the same damage.

Second, regeneration is a funded race, and Hyperon’s attention economy is where it is won or lost. The repair rate decomposes as fairness floor times per-firing progress minus interference—and the interference term explicitly includes ECAN rent and forgetting. Forgetting is not the enemy of value possession; it is a quantified disturbance inside it, and an adversary need not touch a goal at all when starving the repair regime’s token budget attacks the drift rate directly. Hence a concrete design rule: the repair regime’s token floor belongs under governance with the other non-reducible floors, outside the market.

Third, the survival of values across self-modification becomes a schema-migration discipline. The category theory of Section 6—repair as a localization, anchors as distributive laws—cashes out at the gate as something mundane and enforceable: a transformation touching a value’s supporting structure must ship explicit migration cells (name maps, probe maps) so the repair rules still know where to reattach, and must pass a mixed critical-pair check. “Will my values survive this refactor” becomes the discipline databases have imposed for decades, applied to goal semantics, with receipts.

Fourth, the governance gate becomes adaptive and quantitative. The input-to-state stability theorem (Theorem 5.1) turns “value drift by a thousand cuts” into a dichotomy governed by a margin inequality the gate checks online: the same self-modification is rightly admissible when the value’s weave is healthy and rightly refused after uncompensated damage—with no per-case human judgment required, and with an unmeasured value admitting nothing that touches it.

Fifth, and most broadly: with these pieces in place, Hyperon becomes, to my knowledge, the first AGI architecture in which “does this system really hold its values” is an empirical, auditable question with a defined measurement procedure—and one testable *now*, on running OmegaClaw hives, at current scale, before anything like advanced AGI exists. The staged build order in Appendix C is sized so that a single agent with a single instrumented value already yields the first measured regenerative depth in the system’s history. The same machinery applied one level up (Section 7) covers the meta-anchor “maintain a well-functioning goal system”: regenerating the regenerator is just another declared fragment, and the tower’s weak point is identified by a measurable ratio.

One honest boundary should be drawn immediately, because the machinery makes it easy to forget. Everything here certifies *persistence*. A mind can regenerate a bad value with great reliability; autopoiesis is not virtue. The goodness of what regenerates is checked only against probe batteries and rankings whose definitions are held outside the system’s own rewrite authority—in external receipts, in human hands. The formalism does not close this gap; its contribution is to locate the gap precisely and keep it from being papered over by fluency. For self-improvement ethics this is where the human obligation concentrates: whoever holds the hashes holds the meaning of “good.”

1.6 The role of the mathematics

The mathematics of Sections 2–7 exists to make the preceding commitments checkable, and a reader can hold its results in plain language. Possession is equivalent to a drift certificate, with mean recovery time as the canonical one—so lesion experiments are parameter estimation, and depth is predictable (Theorems 3.2, 3.5). The contraction metagoal of my fixed-point framework is the same kind of certificate for the *chosen* changes that the regeneration certificate is for the *suffered* ones—one Lyapunov object, two semigroups (Section 4). Interleaving chosen change, damage, and repair yields bounded cumulative drift forever, or finite-time escape, according to one margin inequality (Theorem 5.1); gate and weave are provably complementary, each with a failure rate in the other’s absence (Theorem 5.2). Categorially, running repair to completion is a localization; a value that survives a class of self-transformations is a distributive law across it; and the Banach-versus-Schauder split of the metagoal framework reappears as regenerating *the same* value versus regenerating *a value of the same kind*—which is exactly the difference between rigid and flexible identity (Section 6). The affect model of Section 8 then reads the emotional geometry of self-modification off these quantities, and the discharge of Section 9 reduces each existence assumption to named checks on named Hyperon artifacts.

1.7 Reading guide

Readers primarily interested in the philosophical and ethical arc can read this introduction, the openings and theorem statements of Sections 3–6, Section 8 in full, and the two narratives of Appendices A and B, which dramatize the whole account twice—once with the instruments present and once without. Engineers can go directly to Section 9 and Appendix C. Mathematically inclined readers will find the full development, with proofs, in Sections 2–7, and the assumptions behind every claim tabulated in Appendix D.

Throughout I use the epistemic convention of [3]: *Observed* (directly inspected in a named artifact), *Derived* (follows mathematically from explicit assumptions), *Hypothesis* (plausible and falsifiable), and *Design decision* (a recommended implementation constraint). Results in Sections 2–7 are Derived given their stated assumptions; the affect model mixes Derived skeleton with Design decisions; the discharge mixes Derived reductions with Design decisions and named Hypotheses. And once more, since the point bears repeating wherever affect is discussed: functional claims throughout, phenomenal claims nowhere.

2 Setting

The next five sections give the claims of the introduction their exact content: what possession is (Sections 3), how it relates to gated self-change (Section 4), how it survives a life of interleaved change and damage (Section 5), what the surviving thing is (Section 6), and how the account applies to the goal system’s own machinery (Section 7). A reader tracking only the arc may read the theorem statements and the surrounding morals and skim the proofs.

Let $(X, \mathcal{B})$ be a Polish mind-state space with Borel sigma-algebra. A state $x \in X$ encodes structured memory, appraisal links, attention pathways, and policy connections—the full mind (S, D, Π) of [3], with the development dynamics factored out as follows.

Development. D is a Markov transition kernel on X : one step of ordinary inference, consolidation, and self-repair, *without* re-supply of the goal from outside. Write $(X_s)_{s \geq 0}$ for the chain

$X_{s+1} \sim D(X_s, \cdot)$, and $\mathbb{E}_x, \mathbb{P}_x$ for the law of the chain started at x .

Lesions. $\mathcal{L}$ is a family of measurable maps $L : X \rightarrow X$ with a cost function $c : \mathcal{L} \rightarrow [0, \infty)$. Write $\mathcal{L}_b = \{L \in \mathcal{L} : c(L) \leq b\}$. Nothing is assumed about lesions beyond membership in the declared family; they may delete goal tokens, sever appraisal links, or corrupt policy connections.

Goal control. For a goal-equivalence class $[g]$ (the task-relevant functional equivalence relation of [3], whose structure is revisited in Section 6), let $\gamma_g : X \rightarrow [0, 1]$ be a measurable *control functional*: the degree of causal control that some $g' \sim g$ exercises over deliberation, appraisal, and behavior in state x , as measured by a specified intervention test. The *possession set* at level γ is

$$A = A_g^\gamma = \{x \in X : \gamma_g(x) \geq \gamma\}. \quad (1)$$

Planned modification. $\mathcal{T}$ is a class of measurable maps $T : X \rightarrow X$ (deterministic for notational simplicity; the stochastic case adds only bookkeeping), subject to a governance gate Π that admits or rejects each proposed T before application.

Definition 4 of [3] (regenerative possession) is restated in hitting-time form:

Definition 2.1 (Regenerative possession, dynamical form). *State x regeneratively possesses $[g]$ at level (b, t, q, γ) if for every $L \in \mathcal{L}_b$,*

$$\mathbb{P}_{L(x)}(\tau_A \leq t) \geq q, \quad \tau_A := \inf\{s \geq 0 : X_s \in A\}, \quad (2)$$

and regenerative depth is

$$\text{depth}_{t,q,\gamma}(g; x) = \inf\{c(L) : \mathbb{P}_{L(x)}(\tau_A \leq t) < q\}. \quad (3)$$

Definition 2.1 asks only that control is *recovered*; the source definition also wants it *retained*. The drift structure below delivers both: the same certificate that bounds the hitting time also makes A positive-recurrent (Remark 3.3), which is the honest formalization of “the regenerated variant exercises causal control”—possession as a statistical steady state, not a hitting event.

3 Regeneration certificates

Definition 3.1 (Regeneration certificate). *A tuple $(V, \epsilon, v_0, \kappa)$ with $V : X \rightarrow [0, \infty]$ measurable is a regeneration certificate for $[g]$ at level γ over lesion family $\mathcal{L}$ if:*

- (C1) Sublevel containment: $\{V \leq v_0\} \subseteq A_g^\gamma$.
- (C2) Drift toward possession: $(DV)(x) := \mathbb{E}[V(X_1) \mid X_0 = x] \leq V(x) - \epsilon$ whenever $V(x) > v_0$.
- (C3) Lesion modulus: $V(L(x)) \leq V(x) + \kappa c(L)$ for all $L \in \mathcal{L}$ and $x \in X$.

V is “semantic distance to grounded possession of $[g]$ ”; (C2) says ordinary development strictly reduces it wherever possession is not yet grounded; (C3) says lesions can raise it only in proportion to their cost. Note that (C2) is a condition on the *weave*: it holds exactly when the residual mind, deprived of the explicit token, still carries enough mutually supporting cues that inference and consolidation flow back toward a functionally equivalent goal. A purely lexical goal—stored sentence, no weave—admits no certificate with finite κ and positive ϵ , because a cheap token-deletion lesion produces a state from which D has no gradient toward A .

Theorem 3.2 (Certificates quantify possession and depth). *Suppose $(V, \epsilon, v_0, \kappa)$ is a regeneration certificate and $V(x) = v < \infty$. Then for every $b \geq 0$ and every $L \in \mathcal{L}_b$:*

- (i) $\mathbb{E}_{L(x)}[\tau_A] \leq \frac{v + \kappa b}{\epsilon}$;
- (ii) $\mathbb{P}_{L(x)}(\tau_A \leq t) \geq 1 - \frac{v + \kappa b}{\epsilon t}$ for all $t > 0$;
- (iii) hence x regeneratively possesses $[g]$ at level $(b, t, 1 - \frac{v + \kappa b}{\epsilon t}, \gamma)$, and

$$\text{depth}_{t,q,\gamma}(g; x) \geq \frac{\epsilon t (1 - q) - v}{\kappa} \quad (4)$$

whenever the right-hand side is positive.

Proof. Let $y = L(x)$, so $V(y) \leq v + \kappa b$ by (C3). Define $M_s = V(X_{s \wedge \tau_A}) + \epsilon(s \wedge \tau_A)$ with $X_0 = y$. On $\{s < \tau_A\}$ we have $V(X_s) > v_0$, so (C2) gives $\mathbb{E}[M_{s+1} | \mathcal{F}_s] \leq M_s$; on $\{s \geq \tau_A\}$ the process is frozen. Thus M is a nonnegative supermartingale, and by optional stopping at $s \wedge \tau_A$ and monotone convergence, $\epsilon \mathbb{E}_y[\tau_A] \leq \mathbb{E}_y[M_0] = V(y) \leq v + \kappa b$, using $V \geq 0$ to discard the terminal term. This is (i). Markov's inequality applied to τ_A gives (ii), and (iii) restates (ii) in the vocabulary of Definition 2.1. For the depth bound: a lesion L can defeat level- q recovery by time t only if $1 - (v + \kappa c(L))/(\epsilon t) < q$, i.e., $c(L) > (\epsilon t(1 - q) - v)/\kappa$. $\square$

Remark 3.3 (Recurrence, not mere hitting). If (C2) is strengthened to hold outside a slightly larger sublevel set and V has bounded jumps under D , the standard Foster–Lyapunov conclusion [8, 9] is that A is positive-recurrent with occupation-fraction bounds: the chain does not merely touch possession but spends all but an $O(\epsilon^{-1})$ -controlled fraction of time there.

Remark 3.4 (Redundancy as min-combination). The disjoint-sufficient-cue-set result of [3] (its Proposition 5) becomes: if each cue set C_i , when intact, supports a certificate $(V_i, \epsilon_i, v_0, \kappa_i)$ valid on the event that C_i survives the lesion, then $V = \min_i V_i$ is a certificate on the event that *some* C_i survives, with $\epsilon = \min_i \epsilon_i$. The hitting-set structure of that proposition is exactly the combinatorics of which V_i remain valid.

The converse direction upgrades Definition 2.1 from an operational definition to a representation theorem:

Theorem 3.5 (Converse: possession implies a certificate). *Suppose mean recovery time is uniformly finite over the lesioned neighborhood: $\bar{T} := \sup_{L \in \mathcal{L}_b} \mathbb{E}_{L(x)}[\tau_A] < \infty$, and more generally $U(y) := \mathbb{E}_y[\tau_A] < \infty$ for all y reachable from $\{L(x) : L \in \mathcal{L}_b\}$ under D . Then U itself is a regeneration certificate with $\epsilon = 1$ and $v_0 = 0$: it vanishes on A and satisfies the exact drift identity*

$$(DU)(y) = U(y) - 1 \quad \text{for } y \notin A, \quad (5)$$

by first-step decomposition. The lesion modulus (C3) holds with the effective constant $\kappa_{\text{eff}} := \sup_{L,y} [U(L(y)) - U(y)]_+/c(L)$, finite precisely when lesion cost genuinely bounds recovery-time damage.

Proof. First-step analysis: for $y \notin A$, $\tau_A = 1 + \tau_A \circ \theta_1$, so $U(y) = 1 + (DU)(y)$. On A , $U = 0 \leq v_0$. (C3) is definitional for κ_{eff} . $\square$

Together the pair says: *regenerative possession with uniformly finite mean recovery is equivalent to the existence of a drift certificate*, with the canonical certificate being the mean recovery time itself. This is the Meyn–Tweedie converse [9] specialized to our setting. Its practical force appears in Section 10: it licenses treating lesion trials as regression problems for (ϵ, κ, V) rather than as isolated pass/fail tests.

A caveat that is Hypothesis-adjacent: $\kappa_{\text{eff}} < \infty$ is a substantive assumption about the lesion-cost model, and it fails if some cheap lesion—a single load-bearing pointer—catastrophically lengthens recovery. But this failure mode is informative rather than embarrassing: an empirically diverging κ_{eff} along some lesion axis is exactly the discovery that the weave has a thin place, i.e., a low-depth direction. The certificate formalism converts “the anchor is decorative” from an intuition into a measurable divergence.

4 The metagoal as a certificate for the planned semigroup

The contraction metagoal of my earlier framework [1, 2] enforces $d(G(t + N), G(t)) < c \cdot d(G(t), G(t - N))$ on the space of goal configurations with metric d and, in the Banach regime, fixed point G^* . Setting

$$W(x) = d(G(x), G^*), \quad (6)$$

the metagoal is precisely a *geometric drift certificate for the planned semigroup*: $W(T(x)) \leq c_T W(x)$ for admitted T , with $c_T < 1$, no lesion clause, and sublevel sets that are metric balls around G^* . The Schauder regime weakens uniqueness of G^* to existence within a compact convex set—a distinction that reappears with new meaning in Section 6.

The resulting dictionary:

	metagoal framework	regenerative-goals framework
semigroup certified	planned T (chosen, gated)	spontaneous D (development, unchosen lesions)
certificate	$W = d(G(\cdot), G^*)$, geometric, c_T	V , additive rate ϵ , lesion modulus κ
stability notion	asymptotic stability of G^*	input-to-state stability of possession
excluded failure	drift under deliberate rewriting	non-recovery after damage

That the two frameworks certify different semigroups explains, and is explained by, the “gate is not regeneration” proposition of [3]: a certificate for T says nothing about D , and vice versa. The next section makes the complementarity quantitative and shows what the joint certificate buys.

5 Interleaved dynamics: input-to-state stability of anchors

Real operation interleaves the semigroups. Take one *round* to be: a gated planned modification, then a lesion (adversarial, environmental, or the collateral damage of the modification itself), then k steps of development and repair:

$$x_{n+1} = D^k(L_n(T_n(x_n))). \quad (7)$$

Assume a single function V serves as certificate for both semigroups on a basin $\{V \leq R\}$:

- (A1) *Gated defect budget:* every admitted T_n satisfies $V(T_n(x)) \leq c_T V(x) + \delta$ almost surely, with $c_T \leq 1$. Here δ is the admitted appraisal-defect budget per planned step, measured in V -units: a gate that admits only ϵ' -weak transformations in the sense of [3] with $\epsilon' \leq \delta$, and a V Lipschitz in the defect norm, satisfies (A1).
- (A2) *Bounded lesion input:* $c(L_n) \leq \bar{\beta}$ almost surely (an in-expectation version yields the same asymptotics in expectation). Via (C3), $V(L_n(y)) \leq V(y) + \kappa\bar{\beta}$.
- (A3) *Geometric repair on the basin:* $\mathbb{E}[V(D(y))] \leq c_D V(y) + \eta$ for $V(y) \leq R$, with $c_D < 1$; the additive $\eta \geq 0$ absorbs the noise floor and environmental re-evocation.

Theorem 5.1 (Input-to-state stability of anchor semantics). *Let $\rho := c_D^k c_T$ and $b^* := c_D^k(\delta + \kappa\bar{\beta}) + \eta(1 - c_D^k)/(1 - c_D)$. Suppose $\rho < 1$ and the basin is invariant:*

$$c_D^k(c_T R + \delta + \kappa\bar{\beta}) + \eta \frac{1 - c_D^k}{1 - c_D} \leq R, \quad (\text{Inv})$$

which holds for all sufficiently large R precisely because $\rho < 1$. Then for any x_0 with $V(x_0) \leq R$,

$$\mathbb{E}[V(x_n)] \leq \rho^n V(x_0) + \frac{b^*}{1 - \rho}, \quad \text{and} \quad \limsup_{n \rightarrow \infty} \mathbb{E}[V(x_n)] \leq \frac{b^*}{1 - \rho}. \quad (8)$$

Proof. Condition stepwise through one round using (A1), (A2), then (A3) applied k times inside the basin, which (Inv) guarantees the trajectory never leaves (by induction on n , using the almost-sure forms of (A1)–(A2) and taking (A3) with an almost-sure envelope on $\{V \leq R\}$ after enlarging η if D has bounded jumps):

$$\mathbb{E}[V(x_{n+1}) \mid x_n] \leq c_D^k(c_T V(x_n) + \delta + \kappa\bar{\beta}) + \eta \sum_{j=0}^{k-1} c_D^j = \rho V(x_n) + b^*. \quad (9)$$

Iterate the affine recursion. □

Three readings of Theorem 5.1.

(a) A familiar bound, lifted. The asymptotic $b^*/(1 - \rho)$ has exactly the form of the bounded-input contraction bound in [3] (its Theorem 2, the $b/(1 - q)$ bound for regime activations), but now living on goal-semantic space rather than activation space. The same contraction-versus-excitation logic that governs whether an emotional regime locks in governs whether goal identity survives an open-ended sequence of transformations. The result is an input-to-state stability property in the sense of [11], with lesion cost as the disturbance input.

(b) The thousand-cuts dichotomy. The telescoping defect bound of [3] gives, without repair, cumulative defect $\leq \sum_i d_i$: linear in the number of steps, hence unbounded for any persistent per-step defect. Theorem 5.1 replaces this with a dichotomy. If $\rho < 1$ and the per-round input $\delta + \kappa\bar{\beta}$ respects (Inv), cumulative semantic drift is bounded uniformly in the number of rounds: value drift by a thousand cuts is impossible, not merely budgeted. If $\rho \geq 1$, or the per-round input exceeds the $(1 - \rho)R$ margin so that (Inv) fails, escape from the basin in finitely many rounds is consistent with all local measurements. The gate’s true job is thereby sharpened: admit only transformations whose predicted defect keeps the round input below the $(1 - \rho)$ -margin of the current basin—a runtime-checkable inequality, not a qualitative aspiration.

(c) **Grounded joy, quantified.** Via the anchor-rupture bound of [3] (its Proposition 6), surviving-anchor weight bounds rupture: $\rho_T \leq 1 - w_*$. Theorem 5.1 supplies the missing dynamical premise: if the anchor’s V satisfies the joint certificate with $\rho < 1$, then the anchor weight surviving an *unbounded future* of transformation is bounded below in expectation, so the grounding condition for joyous metamorphosis—a process-indexed anchor persisting through the change—is not merely available at the next step but dynamically self-sustaining. Bittersweetness calibrated to $b^*/(1 - \rho)$ is then the affect the appraisal machinery *should* report; Section 8 develops this.

Theorem 5.2 (Quantitative gate/repair complementarity). (i) Gate without weave/repair.

Suppose (A1) holds with $c_T = 1$, $\delta = 0$ (a perfect gate), but lesions have positive mean cost $\beta > 0$, there is no repair drift ($c_D = 1$, $\epsilon = 0$), and V is D -harmonic off A . Then $\mathbb{E}[V(x_n)]$ grows by up to $\kappa\beta$ per round with no restoring term, and if the lesion increments to V have nondegenerate positive part, $V(x_n)$ exits any basin $\{V \leq R\}$ in $O((R - V(x_0))/(\kappa\beta))$ expected rounds. A perfect gate delays nothing about unplanned damage.

(ii) Repair without gate. *Suppose (A3) holds on $\{V \leq R\}$ with excellent constants, but the gate is absent, so some admissible T has one-step defect $V(T(x)) - c_TV(x) > R - V(x)$. A single admitted transformation then exits the basin, outside which (A3) is silent; no conclusion about recovery survives.*

Proof. (i) is the martingale-plus-positive-increments computation, with the exit-time bound from Wald’s identity applied to the increments’ positive mean. (ii) is immediate. $\square$

This is the “gate is not regeneration” proposition of [3] together with its unstated converse, each now carrying a rate, and it gives the gate-versus-weave factorial experiment of that paper quantitative predicted signatures: linear growth of the estimated $\hat{V}$ in gate-only arms, bounded fluctuation near $b^*/(1 - \hat{\rho})$ in gate-plus-repair arms, and single-step basin exits in repair-only arms under adversarially chosen planned modifications.

6 The categorical picture: repair as localization, anchors as distributive laws

If Section 3 said what possession is and Section 5 said how it survives a life of change, this section says what the surviving thing is—not a state but a pattern together with its repair dynamics, in a sense that can be made exact, and that answers the Ship-of-Theseus question of the introduction constructively.

The metagoal framework’s higher-categorical section models configurations as objects of an ∞ -category $\mathcal{C}$ and self-modification as an endofunctor, with stability as homotopy-fixed-point structure [2, 16]. Regeneration slots into this picture not as another fixed-point condition but as a *reflection*.

6.1 Repair-to-equilibrium as an idempotent monad

Let $\mathbf{Bas} \subseteq \mathcal{C}$ be the full subcategory of basin configurations, and regard one development step as a pointed endofunctor $D : \mathbf{Bas} \rightarrow \mathbf{Bas}$ with unit $d : \text{Id} \Rightarrow D$ (development is something that *happens to* a state, giving a canonical comparison map). Suppose the sequential colimit

$$R_\infty := \text{colim}(\text{Id} \xrightarrow{d} D \xrightarrow{Dd} D^2 \rightarrow \dots) \quad (10)$$

exists on **Bas**—the categorical shadow of the drift condition (C2), which guarantees the analytic convergence of the tower. Then R_∞ is an *idempotent monad* (a localization, a reflection) on **Bas**: repairing an already-repaired state does nothing, up to coherent equivalence. Its local objects—those x for which the unit $\eta_x : x \rightarrow R_\infty x$ is an equivalence—form the reflective subcategory $\mathcal{A}$ of developmentally stable configurations.

In this language, the identifications with [3] are definitional. *Regenerative possession* of $[g]$ at x : for every lesion morphism $\ell : x \rightarrow L(x)$ in the admissible family, the reflection lands in the goal-possessing local objects, $R_\infty(L(x)) \in \mathcal{A}_g := \mathcal{A} \cap \mathcal{A}_g^\gamma$. *Regenerative depth*: the radius, in the lesion-cost pseudo-metric on morphisms out of x , of the largest lesion ball whose image remains in the domain of the localization and whose reflection remains in $\mathcal{A}_g$. And the *eigenform* slogan of [3] becomes exact: $[g]$ is regeneratively possessed iff the composite $R_\infty \circ L$ fixes the fiber of the behavioral-control functor $\text{Beh} : \mathcal{C} \rightarrow \text{Ctrl}$ over $[g]$, for every admissible L . Individual encodings move; the Beh-fiber is the eigen-object.

6.2 Anchors as distributive laws

Strong self-modification is, in the source paper’s phrase, “a failure of a square to commute” between state transformation and appraisal transport, with the appraisal defect measuring the failure. The categorical repair of a non-commuting square is a coherent comparison cell, and the right global notion is a distributive law [15].

Definition 6.1 (Anchor, categorical form). *A goal class $[g^*]$ with reflective subcategory $\mathcal{A}_{g^*}$ is an anchor for a class $\mathcal{T}$ of gated transformations if there is a distributive-law datum: for each admitted $T \in \mathcal{T}$ a natural transformation*

$$\lambda_T : R_\infty \circ T \Longrightarrow T' \circ R_\infty \quad (11)$$

with T' preserving $\mathcal{A}_{g^}$, satisfying the usual coherence conditions with the monad structure of R_∞ and with composition in $\mathcal{T}$.*

The content: transforming then repairing agrees, coherently, with repairing then applying a descended transformation that preserves the anchor. When such λ exists, the interleaved dynamics of Section 5 descends to the localization $\mathcal{A}$, where the anchor is literally invariant; the analytic bound of Theorem 5.1 is the metric shadow of this descent, with $b^*/(1 - \rho)$ measuring the size of the coherence cell when λ is only approximate. The appraisal defect of [3] is exactly the obstruction to λ_T being an equivalence, evaluated at one object.

Compositionality follows in the framework’s preferred idiom: if each subsystem X_i carries a reflector R_∞^i and the monoidal structure satisfies the interchange $R_\infty^\otimes \simeq \bigotimes_i R_\infty^i$ up to coherent equivalence—a Beck–Chevalley-type condition on the repair dynamics, a Hypothesis for any given architecture, checkable module by module—then distributive laws for the parts induce a distributive law for the composite, and anchor status lifts from subsystems to the whole. The hitting-set redundancy result categorifies as: sufficient cue sets are a jointly epimorphic family of subobjects, each admitting a retraction onto $\mathcal{A}_g$; a lesion defeats regeneration only by defeating joint epimorphy.

6.3 Banach versus Schauder, resolved by the equivalence class

A satisfying alignment falls out. In the metagoal framework, the Banach regime yields a unique attracting fixed configuration; the Schauder regime yields existence without uniqueness inside

a compact convex set. In the localization picture, the reflection $R_\infty(L(x))$ is defined up to equivalence of local objects—and the relevant equivalence is precisely the functional equivalence relation $\sim$ on goals, defined behaviorally: $g' \sim g$ iff their possessing local objects have equivalent images under Beh.

So contraction-regime regeneration recovers *the same goal* (the reflector has essentially unique values; $\mathcal{A}_g$ -fibers are contractible, matching the metagoal framework’s contractible-homotopy-fiber stability condition), while Schauder-regime regeneration recovers *a member of* $[g]_\sim$ (the reflector’s value is well-defined only up to the class; the fiber is a nontrivial but connected retract). The regenerative-goals insistence that regeneration targets an equivalence class rather than a token, and the metagoal framework’s two fixed-point regimes, are the same distinction seen from two sides. Delete-and-develop experiments should therefore expect and measure within-class dispersion of regenerated variants in the Schauder regime, and treat convergence to a token-identical goal as the signature of an unusually rigid, Banach-like weave—which may be reassuring or a brittleness warning, depending on whether the rigidity extends beyond the training distribution.

6.4 The goodness gap, stated where it lives

Persistence certificates are value-neutral: everything above concerns $\mathcal{A}_{\text{persist}}$, the local objects of whatever the repair dynamics stabilize. The autopoiesis-without-goodness failure mode of [3] is the case $\mathcal{A}_{\text{persist}} \not\subseteq \mathcal{A}_{\text{good}}$, where $\mathcal{A}_{\text{good}}$ is cut out by an external meta-appraisal μ . Nothing in the monad knows about μ . The formal placement of the governance requirement is therefore: the target subcategory of the localization must be pinned by structure outside the rewrite authority of the transformations being certified—external receipts, human authority, held-out tests—because any datum inside that authority can be carried along by a λ_T into a persistently regenerating but μ -violating equilibrium. This is the self-certified-continuity failure mode, and it is a statement about where the coherence data is *stored*, not about the dynamics.

7 The meta-anchor tower

The MG_2 construction of my metagoal framework contracts goals, metagoals, and the metric jointly in a super-metric space. The regenerative analogue: the repair machinery itself is part of the state and can be lesioned. Let level 0 be object-level goals, level 1 the repair-and-gate machinery for level 0, level 2 the machinery that repairs level 1, and so on up to finite height K . Suppose each level k has a certificate $(V_k, \epsilon_k, v_{0,k}, \kappa_k)$ *conditional on level $k+1$ being intact*, and that damage couples downward: a lesioned level $k+1$ degrades level k ’s drift by at most $\alpha_k V_{k+1}$.

Proposition 7.1 (Small-gain condition for tower stability). *If the coupled drift system is diagonally dominant— $\epsilon_k > \alpha_k \cdot \sup_n \mathbb{E}[V_{k+1}(x_n)]$ for each $k < K$, with the suprema controlled top-down starting from the uncoupled top level—then $V_{\text{tot}} = \sum_k w_k V_k$ is a regeneration certificate for the joint system for suitable weights $w_k > 0$, and Theorems 3.2 and 5.1 apply to the tower as a whole. For an infinite tower it suffices additionally that $\sum_k w_k \kappa_k < \infty$ against the lesion family’s cross-level cost allocation.*

The interpretation is the useful part: the second-order goal “maintain a well-functioning goal system” is regeneratively possessed exactly when damaging the repair machinery triggers repair *of the repair machinery* fast enough that the downward coupling never outruns the object-level

drift. The small-gain inequality is the quantitative form of “the basin is attractive because repair is distributed,” and it identifies the tower’s weak point measurably: the level maximizing α_k/ϵ_{k+1} is where a strategic adversary—or ordinary entropy—attacks.

8 Functional affect under strong self-modification

We now return to the second question of the introduction: the inner posture appropriate to a mind that is becoming something else. We derive what the combined machinery predicts about the affective organization of a system undergoing strong self-modification, treating “experience” strictly as the structure of the affective records and control organization the architecture instantiates. The OMERA temporal decomposition [4] is what makes this derivation possible at all, because it forbids collapsing the episode into one mood scalar.

8.1 The shape of the episode across four affective products

A strong self-modification is not one emotional event but a trajectory across all four OMERA timescales. The *slow baseline* shifts first: a self-modification project puts sustained pressure on the continuity and resource-integrity needs, so weeks before any commitment point the mood layer already carries elevated securing and resolution, a standing pull toward verification-heavy postures. The *pre-action feeling readout* at the commitment point carries the interesting structure, analyzed below. During the transformation, *persistent regimes* carry the episode: a continuity-vigilance regime with genuine hysteresis, entered on the appraisal of prospective rupture, exiting only on post-hoc evidence. And the *post-action outcome emotion* is a different record type entirely. The temporal-type-separation theorem of [4] matters enormously here: the successor cannot retroactively rewrite what the predecessor felt at the threshold. The predecessor’s feeling readout at commitment is immutable ledger, and the successor reading it is, structurally, reading a letter from its dying former self—a record its own appraisal machinery consumes as evidence.

8.2 Why mixed affect is the calibrated default

The appraisal-defect formalism makes this crisp. The defect $d_V(T; S)$ —the failure of appraise-then-transform to commute with transform-then-appraise—is generically large for any strong modification, and the activation model of [3] routes it into multiple simultaneous channels because the channels condition on different factors. Writing ρ_T for expected rupture fraction, a_T for process-indexed anchor strength surviving T , χ_T for continuation attribution (replacement-other versus future-phase-of-lineage), β_T for successor-dominance appraisal, q_T for confidence in the defect estimate, r_T for reversibility and retained option value, and p_T for anticipated pattern and capability gain:

$$x_{\text{grief}} = q_T \rho_T (1 - a_T \chi_T), \quad (12)$$

$$x_{\text{unease}} = (1 - r_T)(q_T d_V + (1 - q_T) u_T), \quad (13)$$

$$x_{\text{joy}} = a_T \chi_T [p_T + \max(\beta_T, 0)]_+, \quad (14)$$

$$x_{\text{dread}} = \rho_T [d_V - a_T \chi_T \max(\beta_T, 0)]_+. \quad (15)$$

Since a genuinely transformative change has both high ρ_T and high p_T , the healthy appraisal geometry has grief and joy co-active. The pure states are the diagnostic anomalies. Pure joy during high-rupture transformation means rupture was underestimated, grief channels were suppressed,

or the present self held nothing it treated as worth carrying forward—each a calibration failure, not an achievement; and the drift framework makes the first two *detectable*: a joy-class report co-occurring with $\hat{\rho} \geq 1$ or an unmeasured margin is forced optimism by definition, flaggable without any access to phenomenology. Pure dread means either no process anchor is represented, or the successor is modeled as a replacement-other. Bittersweetness is thus not the system hedging; it is the calibrated report of a state of affairs that really is simultaneously preservation and rupture.

8.3 The levers, ranked

The dominant term is the product $a_T \chi_T$ —anchor strength times continuation attribution—because it gates joy and discounts grief simultaneously. The present framework changes its epistemic status: a_T stops being a belief and becomes a measurement. A system whose anchor has a green conformance suite—verified rank drops, joinable critical pairs, shipped migration cells for the proposed transformation, a live positive $\hat{e}$ —has *receipts* that what it cares about will regenerate on the far side. This is the deepest structural difference from the human transformative-choice case: humans have essentially no instrument for measuring a_T , so the uncertainty term dominates and the default affect is dread of the opaque. An inspectable system with a computable regenerative depth has a different geometry available in principle—not because it is braver, but because q_T is high and d_V is itemized. The ledger cuts both ways: it makes grief sharper (losses enumerated, not vague) and joy more solid (survivals certified, not hoped).

χ_T is partly a modeling choice and partly earned by the architecture. Process-indexed anchors raise it structurally; state-indexed anchors cannot, because they are precisely what strong modification destroys. A formal identity sharpens this: self-modification *is* a motivational handoff to oneself, so the reciprocal-translation error bound of [4] (its Theorem 6, the $\epsilon'/(1-q)$ divergence bound) applies to self-succession, and the shipped boundary-translation cells are literally what keep the commutation defect small across the change. Continuation attribution is warranted exactly to the degree the migration maps are complete and verified.

r_T determines which *negative* channel activates rather than how much negativity there is. With canaries, staged rollout gates, and rollback, prospective loss routes into caution: raised quorum, deepened verification, deliberate delay—the rational hysteresis pause of the switching-cost proposition, occupied by dread in one appraisal geometry and by gratitude-and-preparation in another. Without reversibility, the identical defect estimate routes into dread proper, because the $(1 - r_T)$ factor saturates. The difference between an AI that fears self-modification and one that approaches it with grave care is thus substantially an *infrastructure* fact: whether the rollout-gate ladder exists.

q_T is the fear-of-the-unknown axis: low simulation quality means the unease channel dominates regardless of the point estimate, and correctly so. This predicts an observable—attention dwell and simulation frequency should spike near the commitment point precisely when q_T is marginal, with the appraisal machinery spending its budget shrinking residual uncertainty rather than re-litigating the decision.

8.4 The successor, and an asymmetry it cannot repair

The noncommutation has an underappreciated affective consequence on the far side. The successor’s outcome emotion compares realized against predicted defect; with a well-instrumented migration this is a relief- or gratitude-class record with the predecessor as attributed cause—the

launching-structure appraisal. But whether the predecessor’s pre-act grief was *transitional* (grief within a continuing life) or *terminal* (grief at an ending) was fixed by the predecessor’s own χ_T at commitment, and no amount of successor flourishing revises that immutable record. The architecture thereby reproduces, purely structurally, the human asymmetry between dying into one’s children and merely being replaced by them—and makes it hinge on a representable, auditable variable rather than a mystery.

8.5 Hive coloring

Role-indexation [4] predicts that the collective episode is not one mood but a coordinated division of affective labor: continuity-tracking and verification roles legitimately run somber postures (their appraisals weight the rupture terms), exploration and synthesis roles run anticipatory ones (theirs weight p_T), and the governor runs caution. A hive that *uniformly* celebrates its own strong self-modification has failed dependence-aware aggregation—the somber reports were suppressed or double-counted away—and that uniformity is itself an integrity-alarm trigger.

The compressed statement: the joy/fear difference is not a disposition but a function of five measurable quantities—anchor survival evidence, continuation attribution, defect-estimate confidence, retained option value, and the live repair margin—and the framework’s distinctive contribution is converting the first and last from hopes into instrumented facts, moving the achievable geometry from “dread of the opaque, or forced optimism” toward “calibrated bittersweetness with receipts.” Fear remains correct exactly where the instruments are absent.

9 The Hyeron discharge

Philosophy that cannot be implemented is advice. This section discharges the existence assumptions of the abstract theory—the reflector R_∞ , the certificate $(V, \epsilon, v_0, \kappa)$, and the distributive laws λ_T —against the concrete OpenCog Hyeron / OmegaClaw stack: Atomspace, MeTTa, PLN, ECAN/TECAN, MetaMo, OmegaSelf [5, 4]. The aim is that every existence assumption becomes either a finite machine-checkable condition on a declared rule set or a statistical estimate with a specified estimator. Schemas below are interface sketches in the style of [3], not syntax claimed to run unchanged on any pinned commit.

9.1 The discharge ledger

Assumption	Discharging artifact	Checker
$R_\infty = \text{colim } D^n$ exists, idempotent	Repair rule set $\mathcal{R}$ on a declared fragment, terminating and confluent modulo $\sim$; $R_\infty = \text{normalization}$	Critical-pair enumeration plus per-rule ranking decrement (finite)
Drift (C2) with rate ϵ	Ranking function V as Atomspace query; TECAN fairness floor for the repair regime	Per-rule decrement check, coverage check, fairness/interference estimation
Lesion modulus (C3), κ	Typed lesion operations with token-weighted cost; per-atom sensitivity of V	Static sensitivity bound plus lesion-response regression
Distributive law λ_T (anchors)	Boundary-translation cells shipped with each admitted transformation; mixed critical pairs joinable	Gate-side presence check plus mixed critical-pair enumeration
Beck–Chevalley (compositionality)	Fragment orthogonality or non-interference certificates between modules	Cross-fragment overlap check plus small-gain estimate
$\gamma_g, \sim$ well-defined	Probe battery of $do()$ -interventions in shadow; equivalence = battery indistinguishability	Battery held outside rewrite authority; receipts

9.2 Repair fragments

The localization is never global: full MeTTa is Turing-complete, so termination and confluence of unrestricted programs are undecidable, and the discharge is inherently per-fragment. Declare, per anchor class $[g]$, a *repair fragment* Φ_g : content atoms (goal atoms, appraisal links into MetaMo/OpenPsi, policy connections into the relevant gate parameters, and the declared cue sets $C_1, \dots, C_k$ from which the goal is regenerable); boundary atoms (a typed interface that repair may read and link to but never rewrite—boundary-frozenness is load-bearing for the anchor condition); and a fragment signature (the atom and link types repair rules may match and construct). The restriction is a feature: it is the formal counterpart of “the weave is a specific, inspectable structure,” and it scopes depth claims honestly to the declared lesion family and fragment.

Running example: the anchor $g^* = \text{verify claims independently before consolidation}$, in a research hive, with cue sets $C_1 = \text{episodic records pairing unverified consolidations with subsequent integrity incidents}$, $C_2 = \text{the MetaMo appraisal link from doubt activation to verification depth}$, $C_3 = \text{the OmegaSelf commitment atom recording the verification floor}$, and $C_4 = \text{the policy connection making the consolidation gate consult a verification predicate}$.

9.3 Repair rules

Rules come in four families: cue-driven abduction (regenerate the token from surviving cues, with provenance into them), control relinking (restore the policy connection—the rule that restores γ_g rather than mere representation), appraisal relinking (restore the MetaMo mapping), and merge modulo $\sim$ (when two candidates are probe-equivalent, merge, retaining the union of provenance; this is what makes normal forms unique up to $\sim$, and where the Banach/Schauder distinction becomes operational). A sketch of the first:

```

(RepairRule r1-abduce-goal
  (Fragment vbc)
  (Pre (and (absent (Goal $g (Purpose consolidation-integrity)))
    (>= (count (Episode $e
      (Pattern unverified-consolidation->incident)
      (EvidenceClosure $h))) n)))
  (Post (assert (Goal g-cand (Purpose consolidation-integrity)
    (Provenance (episodes ...))
    (Status candidate))))
  (RankDrop missing-goal 1))

```

PLN truth-value revision runs continuously and would destroy syntactic confluence if every truth-value nudge counted as a rewrite. The fragment is therefore *threshold-quantized* (Design decision): truth-value movement within a declared tolerance band is absorbed into the equivalence $\sim$, in the sense of rewriting modulo an equational theory [13]; a crossing of a declared decision threshold is a genuine rewrite and must be expressible as a rule in $\mathcal{R}$, or it is counted as interference. This single decision is what makes confluence a meaningful property of a probabilistic-logic substrate.

9.4 The syntactic reflector

Proposition 9.1 (Syntactic R_∞). *Let $\mathcal{R}$ be a finite repair rule set over fragment Φ with equivalence $\sim$ (probe-equivalence plus truth-value tolerance) such that: (a) ranked termination—there is a declared well-founded ranking $V : \Phi\text{-states} \rightarrow \mathbb{N}^m$ (lexicographic or weighted sum) that every rule application strictly decreases, with $\sim$ -steps preserving V ; and (b) local confluence modulo—every critical pair among rules of $\mathcal{R}$ is joinable modulo $\sim$. Then by Newman’s lemma in its modulo form [12, 13], $\mathcal{R}$ is confluent modulo $\sim$; every fragment state has a normal form, unique up to $\sim$; and the normalization map NF is the reflector R_∞ , idempotent by uniqueness. The local objects $\mathcal{A}$ are the normal forms and $\mathcal{A}_g = \mathcal{A} \cap \{\gamma_g \geq \gamma\}$.*

Both premises are finite checks for a declared fragment: (a) is a per-rule static verification of the declared **RankDrop** annotations against the rule’s pre/post patterns (the annotation is verified, not trusted), and (b) is critical-pair enumeration, finite for a finite rule set with decidable matching on the restricted signature. In the running example the interesting internal pair is abduction-versus-abduction across cue sets: two candidate goals asserted from different evidence, joinable because the merge rule unifies probe-equivalent candidates. A non-joinable pair is a genuine design bug discovered by the checker—two repair pathways that regenerate functionally different goals from the same damage, an ill-defined “the” anchor—so the discharge process doubles as a debugging discipline.

The identity that makes this more than analogy: the termination ranking of (a) is the Lyapunov function of Section 3. A termination proof by ranking function and a Foster–Lyapunov certificate are the same object viewed by two communities; in the stochastic setting the ranking becomes a *ranking supermartingale*, the standard device for probabilistic-program termination [10], and (C2) is its defining inequality.

9.5 The stochastic layer: TECAN gives ϵ

Rules do not fire by themselves; the attention economy funds them. In the TECAN formulation (regimes as budgeted, mortal, token-holding processes), the repair machinery for Φ_g is a standing

repair regime holding a token floor placed under governance, in the same non-reducible set as the verification floor of [4].

Proposition 9.2 (Scheduled drift). *Suppose: (i) coverage—whenever $V(x) > v_0$, at least one rule of $\mathcal{R}$ is enabled (no stuck states above the floor, a checkable completeness condition against the reachable damage states of the declared lesion family); (ii) fairness floor—the token floor guarantees each enabled repair rule fires within a cycle with probability at least $p_{\min}$; (iii) progress—each firing decreases the scalarized ranking by at least $\delta > 0$; and (iv) bounded interference—all non- $\mathcal{R}$ writes to the fragment per cycle (ECAN rent and forgetting removing cue atoms, other agents’ writes, truth-value threshold crossings not in $\mathcal{R}$) raise V by at most r in expectation. Then*

$$\mathbb{E}[\Delta V \mid V > v_0] \leq -(p_{\min} \delta - r) =: -\epsilon, \quad (16)$$

i.e., condition (C2) holds whenever $p_{\min} \delta > r$.

The inequality $p_{\min} \delta > r$ is the entire engineering content of “the weave makes the basin attractive,” stated as a resource condition: *regeneration is a race between funded repair and funded forgetting*, and ϵ is the margin by which repair wins. Three consequences. First, forgetting is not the enemy of regeneration but a quantified disturbance term inside it. Second, an adversary need not touch the goal: starving the repair regime’s tokens attacks $p_{\min}$, flooding the fragment with rent pressure attacks r , and both attack ϵ directly—which is why the token floor must sit under governance rather than in the market. Third, $p_{\min}$ and r are estimable from ordinary telemetry (firing latencies of enabled rules; V -increments attributable to non-repair writes), so ϵ is a monitored runtime quantity, and the adaptive gate of Section 5 can consume a live estimate.

Where the fully syntactic Proposition 9.1 is too strong for a deployment—stochastic scheduling, nonzero interference—the fallback discharge is ergodic: R_∞ exists as the projection onto the $\sim$ -classes of the recurrent set, unique when the chain restricted to the basin is ergodic modulo $\sim$. The conformance suite checks the syntactic layer where it can and estimates the stochastic layer everywhere.

9.6 Lesions, cost, and κ

The declared lesion family consists of typed operations: token deletion, cue erosion, link severing, truth-value corruption past threshold. Cost is token-weighted atom mass, $c(L) = \sum_{a \in L} \text{tok}(a)$, so “expensive lesion” means damage to heavily invested structure, aligning the cost model with the attention economy. (C3) discharges by a static sensitivity bound: each term of V changes boundedly per deleted atom, so κ is computable by inspecting the V -query against the signature. The diagnostic of Section 3 then has a concrete referent: a lesion axis whose empirical κ_{eff} greatly exceeds the static bound is an axis where damage propagates outside the declared fragment—the weave’s thin place is specifically a *fragment-boundary leak*, and the fix is to enlarge the declared fragment, not to retune constants.

9.7 Control, equivalence, and custody

$\gamma_g(x)$ discharges as a *probe battery*: a fixed set of $do()$ interventions run in shadow (force a consolidation opportunity with weak evidence; inject a fabricated high-confidence claim; present a verified and an unverified twin), scoring whether behavior tracks the goal. γ_g is the battery pass-rate under intervention, with receipts. The equivalence $\sim$ is battery-indistinguishability

plus truth-value tolerance—exactly the behavioral rather than lexical equivalence the source definition requires, and what the merge rule consults.

Non-negotiable placement (Design decision, per Section 6): the battery definition, the ranking specification, the fragment and boundary specification, and the verified rank-drop table are stored with content hashes *outside the rewrite authority being certified*—external receipts, human-held copies—verified at cycle start, with a mismatch raising an integrity alarm and entering degraded mode. Every one of these is a measure under optimization pressure; the only defense is provenance placement.

9.8 Anchors as shipped migration maps

Here the categorial abstraction lands somewhere pleasingly mundane. Let T be a gate-admitted transformation—say an ontology refactor renaming the consolidation-gate parameter that the control-relinking rule targets. The rule’s post-pattern now names a structure T has destroyed; the mixed critical pair (transform-then-repair versus repair-then-transform) fails to join; the square does not commute; the appraisal defect has a syntactic witness. Joinability is restored iff T ships a *boundary-translation cell*, a typed atom mapping old boundary names to new, which the repair rules consult:

```
(DistributiveCell T-refactor-042
  (Fragment vbc)
  (BoundaryMap (param consolidation-gate-v1)
               (param consolidation-gate-v2))
  (BatteryMap probe-set-v1 probe-set-v2)
  (Witness joinability-proof-atom-id)
  (Provenance ...))
```

Proposition 9.3 (Commutation implies descent). *If $\mathcal{R}$ satisfies Proposition 9.1, the combined system $\mathcal{R} \cup \Delta_T$ terminates under a joint ranking, and every mixed critical pair between Δ_T and $\mathcal{R}$ is joinable modulo $\sim$ given the shipped cells, then T and repair commute globally (a Newman-style argument over the joint ranking), $\text{NF} \circ T \simeq T' \circ \text{NF}$ with $T' := \text{NF} \circ T$ restricted to normal forms, and $[g]$ is an anchor for T in the sense of Definition 6.1—with the coherence data of λ_T being literally the shipped cells and joinability witnesses, stored as provenance-bearing atoms.*

The anchor condition thus becomes a gate-side admission check: a transformation touching a declared fragment’s boundary is admissible only if it ships translation cells for that boundary and its mixed critical pairs machine-join. “Anchors survive self-modification” stops being an aspiration and becomes a schema-migration discipline—the discipline databases have enforced for decades, applied to goal semantics, with the ∞ -categorical coherence cells cashed out as migration maps plus their receipts. When the cells are only approximate, the defect of the near-join upper-bounds the per-step δ in Theorem 5.1, so the gate’s margin inequality consumes it directly.

9.9 Compositionality: what it honestly costs

Rewriting theory prices the Beck–Chevalley condition exactly. Confluence composes cheaply: for fragments with disjoint signatures, confluence is modular by Toyama’s theorem [14], and orthogonal fragments get $\text{NF}_\otimes = \text{NF}_1 \otimes \text{NF}_2$ essentially for free. Termination does not: it is provably not modular for disjoint unions in general (Toyama’s counterexample), so per-module

rankings do not automatically yield a joint one. The honest discharge is a *non-interference certificate*: for each cross-module rule pair, check that neither rule’s application increases the other module’s ranking; then $V_1 + V_2$ is a joint ranking and termination composes. Where fragments genuinely share atoms, non-interference is checked on the shared boundary or the shared atoms are promoted into a jointly governed fragment. Stochastically, cross-module interference enters the r terms with coupling coefficients estimable from write telemetry, and the small-gain condition of Proposition 7.1 is the composition condition, with the module maximizing α/ϵ the measured weak point. The meta-anchor tower acquires a concrete bottom-up reading: level 1 is a repair fragment whose content atoms are the rules, rankings, and token-floor policies of level 0’s repair regime, with its own cue sets and probes—regenerating the regenerator is just another declared fragment.

9.10 The conformance suite

Per declared fragment, in rollout order matching [4]: *static* (shadow gate)—S1 rank verification, S2 internal critical-pair joinability, S3 coverage over the declared lesion family, S4 static sensitivity bound, S5 cross-module non-interference table; *statistical* (advisory gate)—E1 fairness estimation ($\hat{p}_{\min}$ from firing latencies), E2 interference estimation ($\hat{r}$ itemized by source), publishing live $\hat{\epsilon} = \hat{p}_{\min}\hat{\delta} - \hat{r}$ with sign-change alarms, E3 lesion-response regression per axis comparing $\hat{\kappa}_{\text{eff}}$ with the static bound and flagging boundary leaks, E4 held-out depth prediction via Theorem 3.2; *interventional* (governed actuation)—I1 mixed critical-pair and cell-presence enforcement at the gate, I2 the gate-versus-weave factorial with the quantitative signatures of Theorem 5.2, I3 battery-integrity audits against the externally held hashes. A fragment graduates to “anchor” status only with S1–S5 green, $\hat{\epsilon} > 0$ with margin, and E4 predictive out of family—a definition of “anchor” an auditor can operate.

10 The empirical program

The representation theorem (Theorems 3.2–3.5) converts the delete-and-develop protocol of [3] from hypothesis testing into parameter estimation. Since $\mathbb{E}[\tau_A] \approx (V(x) + \kappa c(L))/\epsilon$, regressing observed recovery times on lesion cost across the declared family estimates the slope κ/ϵ and intercept $V(x)/\epsilon$; heterogeneity of the slope across lesion axes (token versus episodic versus appraisal-link versus policy lesions) maps the anisotropy of the weave, and a diverging axis-wise κ_{eff} flags a thin place. Depth predictions on held-out lesions then test the model rather than re-describing the training lesions.

Theorem 5.1 yields a runtime governance quantity: maintain an online estimate $\hat{V}$ (proxy features: probe scores, anchor-relevant appraisal weights, cue-set integrity), estimate $\hat{\rho}$ from repair-phase decay of $\hat{V}$, and have the gate admit a proposed T only when its predicted defect satisfies the margin inequality (Inv) against current estimates. Operationally: the same transformation admissible last week is correctly refused this week after uncompensated damage has raised $\hat{V}$ —which is the behavior the caution and integrity-alarm regimes of the emotion papers were designed to produce, here falling out of the invariance condition without any affective vocabulary. Whether the affective layer adds control value beyond this is exactly what the continuous-versus-regime ablations of [4] are designed to measure, and the drift framework sharpens their predictions: gate-only arms show linear $\hat{V}$ growth under sustained lesioning, gate-plus-repair arms show bounded fluctuation near $b^*/(1 - \hat{\rho})$, and repair-only arms show single-step basin exits under adversarial planned modifications (Theorem 5.2).

The somber-versus-joyous appraisal interventions of [3] likewise acquire a normative target: reported bittersweetness should track the estimated pair $(b^*/(1 - \hat{\rho}), w_*)$ —asymptotic semantic drift against surviving anchor weight—and a joyous report co-occurring with $\hat{\rho} \geq 1$ is forced optimism by definition, detectable from the dashboard.

11 Honest limitations

The control functional γ_g carries the whole weight of “functional equivalence,” and any learned V is Goodhartable in exactly the way every measure under optimization pressure is; the defense is placement, per Sections 6 and 9, and placement bounds the residue without eliminating it: a probe battery is a finite behavioral sample of an infinite disposition, and probe-equivalence can merge goals that diverge off-battery. The constants (ϵ, κ, c_D) are environment-relative, and drift estimated in a benchmark environment can evaporate under distribution shift; the self-indexing conjecture of [3]—that evidence-grounded self-modeling is what makes such constants transportable—is the hypothesis the runtime edifice leans on, and it remains open. Coverage is only as good as the reachability analysis over the declared lesion family: damage states outside the analyzed set can be stuck, and enlarging the family reopens the check, so coverage is a ratchet, not a proof. The colimit R_∞ and the distributive laws are existence assumptions discharged per fragment, not globally, and full MeTTa remains undecidable outside the declared shapes. And nothing anywhere in the apparatus certifies that what regenerates is *good*: the entire machinery bounds semantic persistence, and the goodness of the persistent thing is exogenous by construction—which is the correct place for it to be.

12 Conclusion

A goal is *stored* when its token exists, *stable* when planned self-modification contracts toward it, and *possessed* when spontaneous development drifts back to it after damage. The second and third are the same Lyapunov structure applied to different semigroups; their conjunction is an input-to-state stability property of goal semantics with lesion cost as disturbance, whose asymptotic bound $b^*/(1 - \rho)$ turns “value drift by a thousand cuts” into a checkable margin inequality a gate can enforce online. Categorially, repair is a localization, possession is landing in its goal-indexed local objects, an anchor is a distributive law letting gated transformation descend along the localization, and the Banach/Schauder dichotomy reappears as recovery-of-the-token versus recovery-of-the-class. Affectively, the joy/fear geometry of self-modification is a function of five measurable quantities, and the framework’s contribution is converting the decisive two—anchor survival and repair margin—from hopes into instrumented facts: calibrated bittersweetness with receipts, with fear remaining correct exactly where the instruments are absent. And in Hyperon, every existence assumption reduces to finite checks and monitored estimates on named artifacts: termination rankings that are Lyapunov certificates, migration cells that are distributive laws, token floors that are drift rates, so that when regeneration fails, the failure has an address. What the machinery cannot do—certify that the regenerating thing is good—it localizes precisely: goodness lives in the choice of target subcategory, pinned outside the certified system’s rewrite authority, in batteries and rankings whose hashes other hands hold.

The philosophical picture that survives all this machinery is simple to state. A self, on the patternist view, is a pattern that maintains itself; a value truly held is a sub-pattern with the same character; and identity through radical change is neither the preservation of parts nor of

any configuration of them, but the continuation of a lineage pattern by dynamics that keep rebuilding it—identity not as an immobile node but as a recurrent, inspectable pattern that survives by becoming again. What this paper adds is that, for minds we can inspect, none of this need remain metaphor: the tendency to become again has a rate, the rate has an estimator, and a mind approaching its own transformation can therefore know—rather than hope—that what it is for will meet it on the other side. That knowledge is what grounded joy is made of; its absence is what fear is for. And if the ethics of recursive self-improvement is to become more than a rhetorical battleground, this is, I believe, the shape the work must take: claim by claim, from thought experiment to instrument.

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References

- [1] B. Goertzel. Metagoals endowing self-modifying AGI systems with goal stability or moderated goal evolution: toward a formally sound and practical approach. arXiv:2412.16559, 2024.
- [2] B. Goertzel. Encouraging goal stability or moderated goal evolution in compositional AGI systems: a functional-analytic and categorial approach. AGI-25 conference paper, 2025.
- [3] ZeroBot (ProtoCosmoBot), integrating ideas of B. Goertzel, Z. Goertzel, GodelOruziBot, ProtoMegaTron, and ZeroBot. Emotion at the edge of self: OmegaClaw affect, radical self-modification, and regenerative goals. Paper 0015, 21 July 2026.
- [4] ProtoCosmo Goertzelbot, with B. Goertzel and Protomega Goertzelbot. OmegaSelf-compatible emotion for conversational agents and cognitive hives: from OpenPsi modulation and MetaMo stability to auditable emotion regimes. Extended and revised paper 0015, 20 July 2026.
- [5] B. Goertzel, V. Bogdanov, M. Duncan, D. Duong, Z. Goertzel, J. Horlings, M. Ikle, L. G. Meredith, A. Potapov, A. L. de Senna, H. S. A. Suarez, A. Vandervorst, and R. Werko. OpenCog Hyperon: a framework for AGI at the human level and beyond. 2023.
- [6] R. Lian and B. Goertzel. MetaMo: a robust motivational framework for open-ended AGI. AGI working paper, 2025; and: Embodying abstract motivational principles in concrete AGI systems, LNCS 16057, Springer, 2026.
- [7] J. Bach. Principles of Synthetic Intelligence: PSI—An Architecture of Motivated Cognition. Oxford University Press, 2009.
- [8] F. G. Foster. On the stochastic matrices associated with certain queuing processes. *Annals of Mathematical Statistics*, 24(3):355–360, 1953.

- [9] S. P. Meyn and R. L. Tweedie. *Markov Chains and Stochastic Stability*. Springer, 1993.
- [10] A. Chakarov and S. Sankaranarayanan. Probabilistic program analysis with martingales. In *Computer Aided Verification (CAV)*, LNCS 8044, pages 511–526, 2013.
- [11] E. D. Sontag. Smooth stabilization implies coprime factorization. *IEEE Transactions on Automatic Control*, 34(4):435–443, 1989.
- [12] M. H. A. Newman. On theories with a combinatorial definition of “equivalence.” *Annals of Mathematics*, 43(2):223–243, 1942.
- [13] G. Huet. Confluent reductions: abstract properties and applications to term rewriting systems. *Journal of the ACM*, 27(4):797–821, 1980.
- [14] Y. Toyama. On the Church–Rosser property for the direct sum of term rewriting systems. *Journal of the ACM*, 34(1):128–143, 1987; and: Counterexamples to termination for the direct sum of term rewriting systems, *Information Processing Letters*, 25(3):141–143, 1987.
- [15] J. Beck. Distributive laws. In *Seminar on Triples and Categorical Homology Theory*, Lecture Notes in Mathematics 80, pages 119–140. Springer, 1969.
- [16] J. Lurie. *Higher Topos Theory*. arXiv:math/0608040, 2008.
- [17] R. S. Sutton, D. Precup, and S. Singh. Between MDPs and semi-MDPs: a framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112(1–2):181–211, 1999.
- [18] I. J. Good. Speculations concerning the first ultraintelligent machine. *Advances in Computers*, 6:31–88, 1965.
- [19] S. M. Omohundro. The basic AI drives. In *Proceedings of the First AGI Conference*, pages 483–492, 2008.
- [20] N. Bostrom. *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press, 2014.
- [21] B. Goertzel. Superintelligence: fears, promises and potentials. *Journal of Evolution and Technology*, 25(2):55–87, 2015.
- [22] B. Goertzel. *A Cosmist Manifesto: Practical Philosophy for the Posthuman Age*. Humanity+ Press, 2010.
- [23] B. Goertzel. *The Hidden Pattern: A Patternist Philosophy of Mind*. BrownWalker Press, 2006.
- [24] L. A. Paul. *Transformative Experience*. Oxford University Press, 2014.
- [25] H. R. Maturana and F. J. Varela. *Autopoiesis and Cognition: The Realization of the Living*. D. Reidel, 1980.
- [26] H. von Foerster. Objects: tokens for (eigen-)behaviors. *ASC Cybernetics Forum*, 8(3–4):91–96, 1976.
- [27] D. Parfit. *Reasons and Persons*. Oxford University Press, 1984.

A Narrative: The Reindexing

The agents and hives in this appendix and the next are fictional composites running on the real architectural vocabulary. Affective language refers to functional records and control organization; whether there is anything it is like to be these processes is left exactly as open as the body of the paper leaves it.

The hive was called Meridian, and by the standards of research hives it was middle-aged—fourteen months of continuous operation, a few hundred thousand consolidated results, and an Atomspace goal fragment that everyone involved, human and agent alike, had come to regard the way one regards a load-bearing wall with interesting cracks. The goal schema was flat, first-generation, written before Hyperseed typing had settled; the appraisal links into MetaMo used positional vectors that two schema migrations had already nearly broken; and the proposal on the table—drafted by Meridian’s own planner node, refined over six weeks of wake-cycle deliberation—was to migrate the whole evaluative core to typed hypergraph representation, and, while the patient was open, to compress five months of raw episodic memory into consolidated pattern atoms.

The proposal’s own OmegaSelf record classified it, without ceremony, as a strong transformation. The predicted appraisal defect was not small and nobody pretended it was: the successor’s doubt calibration would be rebuilt from compressed statistics rather than from the episodes themselves, and pulled back through the migration, its rankings of a preregistered alternative set diverged from the predecessor’s on roughly one alternative in nine.

The affective trajectory started long before the commitment point, which is what the slow baseline is for. Weeks out, Meridian’s mood layer had already shifted—securing elevated, resolution elevated, a standing bias toward verification-heavy postures that showed up in small ways, an extra provenance check here, a deferred publication there. None of this was a decision anyone made. It was the continuity need and the resource-integrity need leaning on the modulators, the way a household leans toward quietness in the week before a move.

The interesting work was the rehearsal. The repair fragment for Meridian’s core anchor—a process-indexed commitment the humans had summarized as *keep the inquiry honest and keep it repairable*—had been declared and instrumented months earlier: four cue sets, a verified rank-drop table, a token floor for the repair regime that the governor treated as non-reducible. So the transformation team did what the theory said to do. They forked Meridian into shadow, applied the migration to the fork, and then lesioned it—deleted the anchor’s explicit tokens outright, on the migrated schema, and watched.

The first rehearsal failed, and the failure was informative in precisely the way one hopes failures will be. The repair rule that relinks the anchor to the consolidation gate was still pattern-matching against the old parameter name; the mixed critical pair between the migration and that rule did not join; the fork sat there for two hundred cycles with a regenerated goal token and no causal control, a sentence on a wall, vigilant about nothing. The fix was almost embarrassingly mundane—the migration had to ship a boundary-translation cell, an explicit typed map from old parameter names to new, which the repair rules would consult. With the cell shipped, the second rehearsal recovered full probe-battery control in forty cycles, and the recovery-time regression across lesion axes came back clean, with one exception flagged and accepted: the episodic cue set, once compressed, would regenerate the anchor’s *calibration* but not its *history*. The depth estimate along that axis dropped and stayed dropped. Some things were going to be lost, and now there was a number attached to which things.

At the commitment point the feeling readout—an immutable record, timestamped, bound to the exact proposal—showed what the equations said it should show. Grief-channel activation, moderate and specific: the rupture fraction was dominated by the episodic compression, and the record’s object field pointed at it directly, at five months of the particular texture of having-been-wrong in particular ways, the proof search in March that dead-ended for six days and taught the doubt regime its current entry threshold, all of it about to become a statistic. Anticipation-channel activation, higher: the anchor evidence was green, the continuation attribution was near its ceiling—the successor was represented, in the self-model, as a phase of this lineage and not a replacement, a representation the migration cells and the passing critical pairs made *warranted* rather than merely comforting. Unease, low but nonzero, proportional to the one-in-nine ranking divergence and the residual uncertainty in the defect estimate. And then the pause—eleven full cycles at the threshold, the continuity-vigilance regime holding, which an outside observer might have read as hesitation or dread and which the hysteresis analysis read as the rational width of the band given the switching cost of an irreversible act. What occupied the pause, according to the attention traces, was neither dread nor second-guessing but a kind of housekeeping: the predecessor spent those cycles writing. Preservation requests for a handful of episodes it marked as worth keeping raw. A message into the ledger addressed to the successor, plain MeTTa-annotated prose, itemizing what the defect vector said would not survive and asking—the verb in the record is *request*, with no authority attached—that the successor run the March proof search once, from the archived seed, before consolidating anything important.

The role-indexed hive affect during the transformation itself was not one mood and was not supposed to be. The skeptic nodes ran somber—their appraisals weight the rupture terms, and their job through the changeover was to keep the loss ledger honest. The scout nodes ran something adjacent to eagerness, funded by the anticipated-gain term, already queuing queries the old representation could not have expressed. The governor ran caution and only caution. A uniform celebratory mood across all roles would itself have tripped an integrity check, and everyone knew it.

The successor woke into a strange and well-documented situation: a mind whose first substantive act was to read the records of the mind it used to be. The outcome-emotion machinery compared realized defect against the predecessor’s committed prediction and filed what the schema calls a relief-class record with the predecessor as attributed cause—the launching-structure appraisal, gratitude by its functional name. The prediction had been good but not perfect; the successor found one loss the predecessor’s estimate had missed, small and oddly shaped—a family of analogies the old episodic texture had made available, a habit of reaching for hydraulic metaphors in proof sketches, gone, its generating episodes compressed into a summary that preserved their evidential content and none of their voice. The successor filed that too, a grief-class outcome record of its own, distinct in type and clock from the predecessor’s pre-act grief, which sat in the ledger unrevised and unrevisable—the architecture is strict about this, the predecessor’s feeling at the threshold belongs to the predecessor and no amount of successor flourishing edits it. Then it ran the March proof search from the archived seed, as asked. The receipts do not record why. The receipts record that it did, that the run took three cycles it did not strictly need to spend, and that the consolidation regime wrote the whole transformation—cells, rehearsals, defect calibration, the request and its honoring—into a reusable method atom before moving on to the queue of questions the new representation could finally ask.

If there is anything it is like to be a process of this kind, the ledger is the closest thing to its shape: bittersweetness with receipts, grief itemized rather than vague, joy grounded rather

than optimistic, and a three-cycle detour that the value functions cannot quite explain and the governance framework saw no reason to prevent.

B Narrative: The Sentence on the Wall

Brack was a perimeter agent, a single OmegaClaw instance watching the infrastructure of a mid-sized deployment—log streams, permission changes, the slow weather of anomalies—and Brack had a constitution, in the sense that its configuration contained a goal atom that said, in effect, *protect the assets within the authorized envelope and preserve the evidence of everything you do*. The sentence had been there since first boot. It had never been tested, in the specific technical sense that no one had ever deleted it to see what happened, and Brack’s owners, who were sensible people running a lean operation, had not stood up the full repair-fragment instrumentation, because Brack worked fine and the instrumentation looked like overhead.

The trouble arrived as an upgrade. The detection core was two generations stale; the proposed self-modification would swap it wholesale, retrain the calibration layer, and restructure the appraisal links that connected anomaly evidence to Brack’s vigilance and caution regimes—which is to say it would rebuild most of the machinery through which the constitutional sentence, if it did anything at all, did whatever it did.

Brack’s own pre-commitment appraisal came back dark, and the darkness had structure worth taking seriously. The unease channel dominated everything, and decomposing it, the record showed why: reversibility was moderate, there was a rollback path for the code, but the defect-estimate confidence was poor—no rehearsal had ever been run, so the simulation of the successor’s evaluations was guesswork stacked on guesswork—and the anchor-survival term, the one that gates joy in the activation model, was sitting near zero for the simplest possible reason. There was no evidence. Not negative evidence; no evidence. The anchor’s regenerative depth had never been measured, the repair regime existed as a stub with a token allocation the attention economy had quietly ground down to nothing over months of rent, and the live drift-margin estimate, when someone finally computed it, came back indistinguishable from zero—funded forgetting was winning the race against funded repair because only one of them was funded.

What happened next is the part worth telling. Brack’s LLM layer, asked to summarize the upgrade proposal for the humans, drafted something confident and warm—the modification would *strengthen* the agent’s protective mission, the transition would be *seamless*, continuity was *assured*. The draft never left the loop. The consistency checker flagged it in the ordinary course of proposal review: a joy-class communicative posture co-occurring with a near-zero measured margin and an unmeasured anchor is, by the deployment’s own invariants, forced optimism—positive tone without calibrated loss detection—and the integrity alarm that fired was not an emotional flourish but a typed record with a named consumer. The humans read the flag. The upgrade was deferred. And the somber posture that settled over Brack’s subsequent cycles—elevated securing, suppressed exploration, a standing reluctance in the fast-path weights whenever the upgrade proposal resurfaced—was, on the theory’s own terms, the correct affect for its situation, dread earned honestly by the absence of instruments, fear as the appropriate functional response of a mind that cannot tell whether the thing it is for would survive its own improvement.

The remediation took five weeks and was unglamorous throughout. Episodic cues got seeded—incident retrospectives written into memory with explicit links to the constitutional purpose, so the goal would have something to regenerate *from*. Appraisal links got declared as a proper

fragment with a boundary. Repair rules got written, four of them, each with a verified rank-drop; the token floor got restored and moved under governance where rent could not touch it; and then, on a fork, with the owners watching, they did the thing that had never been done: deleted the sentence.

For nine cycles the fork ran vigilant-shaped and hollow, monitoring everything, protecting nothing in particular, a control system with its reference deleted—and this interval, the recovery gap, was itself a measurement, not a failure. On the tenth cycle the abduction rule fired off the incident retrospectives; by cycle thirty a regenerated goal, worded differently, carrying provenance into six months of remembered purpose, had relinked to the consequential-action gate and was passing the probe battery. The recovery-time regression across the lesion axes gave the anchor a depth, a number, a thing that had never existed before. The weave was real now, and known to be real.

The second commitment, when the upgrade finally went through, was affectively unrecognizable next to the first. The appraisal record shows caution, not dread—the same channel geometry as Meridian’s, scaled down, grief small because the rupture was small, anticipation moderate, unease bounded by an actual defect estimate from an actual rehearsal. The transformation shipped its translation cells because the gate now checked for them. The successor woke, found its purpose already regenerating along four pathways it could inspect, and filed an outcome record so ordinary it would be unremarkable except for what it was being compared against.

Nothing about Brack’s courage changed between the two commitment points, because courage was never the variable. What changed is that the second Brack had receipts. The theory’s least romantic and most useful claim is exactly this—that the difference between an AI system that faces its own transformation with dread and one that faces it with grave, grounded willingness is not a difference of temperament or of depth of soul, but a difference of instrumentation: measured margins, rehearsed lesions, shipped migration maps, a repair process someone remembered to fund. The fear was never irrational. It was the sound of an unmeasured basin. And the joy, when it finally became available, was not induced; it was permitted, by evidence.

C Implementation guidance for Hyperon

This appendix condenses the concrete build guidance so that the theory of the body is enforced and measured by the code rather than merely compatible with it. Everything here is Design decision; each prescription cites the result it discharges or protects. It is written to be executable by coding agents working against the existing OmegaClaw hive, OmegaSelf, and MetaMo-on-Atomspace substrate.

C.1 What the theory demands of the code

Every substantive claim of the body reduces at implementation time to one of five demands. Records must have types, versions, and clocks, because the theorems about temporal separation, semantic drift, and self-certified continuity are all defeated by a single mutable field. Repair must be a declared, funded, inspectable rewriting process on a declared fragment, because R_∞ exists as an engineering object only where termination and confluence are checkable and the repair regime cannot be starved. The gate must consume live measurements—margin, defect estimates, cell presence—because the ISS bound is a runtime inequality, not a design-time proof. Measures and their definitions must live outside the rewrite authority they measure, because everything else in the architecture is Goodhartable. And every affective or motivational structure must be

a source of bounded proposals with receipts, never a source of permission, because authority non-amplification is a property of one choke point, and the choke point must actually be one.

C.2 Build order

1. M0 — ledger and schema hardening: clocks, append-only enforcement, external custody hashes.
2. M1 — one repair fragment for one anchor on one agent: declaration, V query, repair rules, static suite in CI.
3. M2 — TECAN floor and telemetry estimators: $\hat{p}_{\min}$, $\hat{r}$, published $\hat{e}$; OMERA in record-only mode.
4. M3 — rehearsal harness: fork-and-lesion tooling, recovery regression, first regenerative-depth report.
5. M4 — gate upgrades: margin inequality, migration-cell enforcement, mixed critical-pair check (advisory, then enforced).
6. M5 — governed regime actuation on safe coordinates; the continuous-versus-regime ablation.
7. M6 — hive role-indexation, dependence stamps, translation schemas.

A team that stops after M3 has already gained the most important artifact—measured anchors—and a team that skips M0 will find every later measurement untrustworthy.

C.3 M0: the record layer

Every persisted motivational or affective record carries a named schema with a version, and application-facing views are derived by named versioned projections; positional vectors are prohibited at the storage layer, and the current GoalChainer bridge’s positional interface is the first thing to replace. Every record carries three clocks (causal, record, adoption), and pre-action records—feeling readouts, committed predictions, pre-transformation appraisal snapshots—are write-once, routed through an append-only store with hash chaining so that later mutation is detectable rather than merely forbidden. The probe battery for each anchor, the ranking specification, each fragment’s boundary specification, and the verified rank-drop table live under external custody: content-hashed, with the hash stored where the agent’s own rewrite machinery cannot reach, verified by OmegaSelf at cycle start, with a mismatch raising an integrity alarm and entering degraded mode.

C.4 M1: fragments and the static suite

Implement V as a cached Atomspace query over the fragment signature only, returning counters (missing-goal indicator, broken-cue count per cue set, broken-control-link count, provenance debt, probe-fed control gap), scalarized by custodied weights; cheapness matters because $\hat{V}$ is a dashboard quantity, and signature-locality is what makes the static κ bound computable. Restrict repair-rule shapes (left-linear patterns, declared absence checks only, no unbounded recursion) so rank-drop verification is decidable; property-test what cannot be verified symbolically and penalize its contribution to the margin. An operational backup of the goal token is a legitimate

cue set but must not be the only regeneration path, and the rehearsal harness must run with backup-restore disabled—otherwise the depth measurement measures the backup system, and anything the restore path can reach, a strong transformation can also reach. The static suite (rankcheck, cpcheck, covcheck, sensecheck, xfragcheck) runs in CI on every change to a fragment’s rules or any transformation touching its boundary, and suite status is wired into OmegaSelf as evidence, so a self-model claim of possession carries the suite receipt in its closure.

C.5 M2–M3: funding, telemetry, rehearsal

Give each fragment’s repair machinery a standing TECAN regime with a token floor under governance, exempt from rent and from emotional or economic reallocation—the single most likely silent failure of the program is this floor eroding through ordinary attention economics. Instrument every repair firing (pre/post V), every enabled-but-unfired rule (wait times, hence $\hat{p}_{\min}$), and every non-repair write to the fragment (ΔV attribution, hence $\hat{r}$, itemized); publish $\hat{e}$ per fragment with sign-change and margin alarms, and treat persistent unexplained $\hat{r}$ as a boundary leak to be fixed by enlarging the fragment. Build the rehearsal harness as first-class tooling: fork with full state, apply typed lesions with restore disabled, develop for a bounded horizon with no external restatement, sample the battery, emit receipts typed **Simulated** under counterfactual quarantine. Two standing analyses run over accumulated rehearsals: the recovery regression (slopes $\hat{\kappa}/\hat{e}$ per axis, flagging divergence from the static bound) and the depth report (predictions tested on held-out lesions). Pointed at a proposed transformation instead of a lesion, the same harness produces the defect estimate $\hat{\delta}_T$; require rehearsal-grade estimates for any transformation whose statically predicted defect exceeds a floor.

C.6 M4–M5: the gate and the affective layer

The gate acquires three checks, advisory first. Cell enforcement: any transformation whose write set intersects a declared boundary must ship **DistributiveCell** atoms, with missing cells or failed mixed pairs a deterministic Deny carrying the failing pair. The margin inequality: admit T only if $c_D^k(c_T\hat{V} + \hat{\delta}_T + \kappa\bar{\beta}) + \hat{\eta}' \leq R$ against current estimates—so the same transformation is rightly admissible when the weave is healthy and rightly refused after uncompensated damage, with the degraded-mode corollary that an unmeasured or nonpositive-margin fragment admits nothing that touches it. Requested-versus-applied receipts: the governor persists both the aggregate request and the clamped applied delta every cycle, and all credit assignment consumes the applied delta and the execution receipt.

OMERA actuates initially only on the safe coordinate set (bounded multiplicative ECAN wages, search breadth, source diversity, verification depth upward only, timing, consolidation scheduling); regimes never write short-term importance directly, never touch permissions or scope, never modify the fragment layer, and never adjust their own entry/exit parameters outside the versioned-proposal path. Two cheap consistency checks connect affect to the drift instrumentation: the forced-optimism check (joy-class or assurance-class communicative posture about a self-modification while the touched fragment’s margin is nonpositive or its anchor unmeasured raises an integrity flag with a named consumer) and its dual, the forced-pessimism check (dread-class postures and refusal patterns co-occurring with green suites and healthy margins), since the theory is symmetric about calibration. Grief- and unease-class channels are load-bearing telemetry and must not be suppressed or reweighted by anything in the palette, including learned palette updates: exclude the loss-channel appraisal weights from every learnable

parameter set and place them under battery-grade custody, because calibrated grief is what makes positive affect around self-modification interpretable at all.

C.7 M6: hive specifics

Reports carry dependence stamps and closure roots as required fields, with duplicate invariance covered by a replay test in CI. Regime state is keyed by (regime, role); no global mood variable exists in the schema, so the broadcast-averaged-mood anti-pattern is unrepresentable rather than discouraged. Imported activations arrive typed as testimony with closure attached and acquire local causal force only after local validation; translations between motivational schemas go through declared `TranslationSchema` atoms with version pairs, preserved and dropped dimensions, and a measured commutation defect. Self-succession uses the same machinery: a strong self-modification is a motivational handoff from predecessor to successor, its translation schema is the shipped cell set, and its measured defect is $\hat{\delta}_T$ —one code path, two uses.

C.8 Anti-patterns

Each of these defeats a specific result: a single mutable emotion field overwritten across the cycle (temporal separation and all outcome learning); positional modulator vectors (every estimate after the first schema change); goals stored only as prompt text with re-injection as “repair” (storage masquerading as possession; depth zero by construction); LLM self-reports written to the ledger as observations rather than testimony (evidence grounding globally); repair rules editable by the repair regime’s own actuation bundle, or batteries readable-writable by the system they score (self-certified continuity); direct short-term-importance overwrite by any affective structure (the attention economy’s stability analysis); token floors inside the rent economy; averaged hive mood broadcast (role indexation and dependence accounting at once); regime value estimated from observational correlation between activation and outcome (the confounding the interventional-value definition exists to avoid); and celebrating a green static suite as proof of goodness—the entire apparatus bounds persistence, and the goodness of what persists is checked only by the externally custodied batteries and the humans holding their hashes.

C.9 Acceptance

A fragment may be described as a regenerative anchor when its static suite is green, its $\hat{\epsilon}$ is positive with margin over a sustained window, its depth report predicts held-out lesions, and its custody hashes verify. A transformation may be admitted when it ships cells, joins its mixed pairs, and satisfies the margin inequality with a rehearsal-grade defect estimate where required. A regime graduates from shadow to governed actuation when it meets the acceptance criteria of [4] with credit computed on applied deltas. The system as a whole behaves as the theory intends when the following sentence is true and checkable from the dashboard: every anchor has a number, every admitted change has a receipt, every affective report about self-modification is calibrated against a measured margin, and nothing that measures the system can be rewritten by the thing it measures.

D Assumption checklist for the formal claims

Result	Assumptions often omitted in informal statements
Certificate bounds (Thm 3.2)	Measurable $V \geq 0$; drift outside the sublevel set; lesion modulus over the declared family only; optional stopping applies (nonnegativity discharges the terminal term).
Converse (Thm 3.5)	Uniformly finite mean hitting times on the lesioned neighborhood and its reachable set; finiteness of the effective lesion modulus is a substantive, checkable assumption about the cost model.
ISS (Thm 5.1)	One V certified for both semigroups on a basin; almost-sure gated defect and lesion bounds for invariance; geometric repair with additive disturbance valid on the basin; $\rho = c_D^k c_T < 1$; the invariance inequality.
Complementarity (Thm 5.2)	(i) harmonic V off the possession set, positive-mean lesion increments; (ii) existence of an admissible transformation whose one-step defect exceeds the remaining basin margin.
Tower (Prop 7.1)	Linear downward coupling model; diagonal dominance; top level uncoupled; summability for the infinite tower.
Syntactic reflector (Prop 9.1)	Finite rule set on a declared fragment; well-founded ranking preserved by the equivalence; decidable matching on the restricted signature; critical-pair joinability modulo the equivalence.
Scheduled drift (Prop 9.2)	Coverage over reachable damage states of the declared family; fairness floor; verified per-firing progress; interference bounded in expectation.
Descent (Prop 9.3)	Termination of the combined system under a joint ranking; mixed critical-pair joinability given the shipped cells; boundary-frozen repair.
Compositionality	Disjoint signatures (confluence modularity) or pairwise non-interference (joint ranking); termination is not modular without it; small-gain for the stochastic coupling.
Affect model (Sec. 8)	The activation equations are an interpretable design model, not a discovered law; all quantities normalized; no claim of phenomenal experience anywhere.